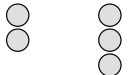
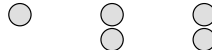
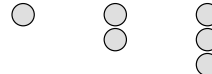
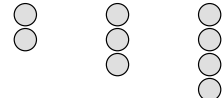


Take turns. Remove one or more counters from one pile. Whoever takes the final counter loses.

Problem 1: Build these starts with counters. Which player can always win? Find a rule for all two-pile starts with at least one counter.

Start 1  Winner: _____ 1 0	Start 2  Winner: _____ 1 1	Start 3  Winner: _____ 1 2	Start 4  Winner: _____ 2 2
Start 5  Winner: _____ 2 3	Start 6  Winner: _____ 3 3	Start 7  Winner: _____ 2 0	Start 8  Winner: _____ 1 1 1
Start 9  Winner: _____ 1 1 2	Start 10  Winner: _____ 1 2 2	Start 11  Winner: _____ 1 2 3	Start 12  Winner: _____ 2 3 4

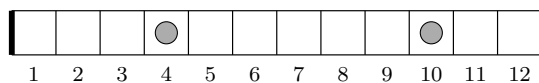
Play here

Pile 1	Pile 2	Pile 3
---------------	---------------	---------------

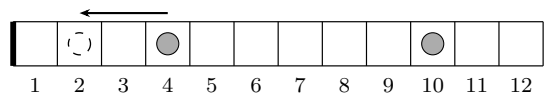
Rule: _____

Take turns sliding one coin left through empty squares. Coins cannot jump. A player with no move loses.

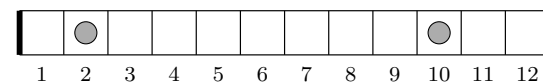
Before



Slide the coin from 4 to 2

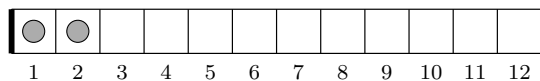


After



Problem 2: Play these starts. Which player can always win?

Start 1



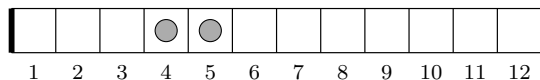
Winner:

Start 2



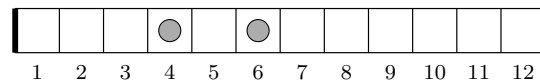
Winner:

Start 3



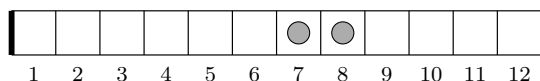
Winner:

Start 4



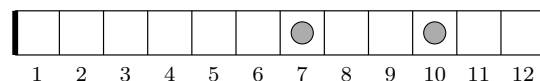
Winner:

Start 5



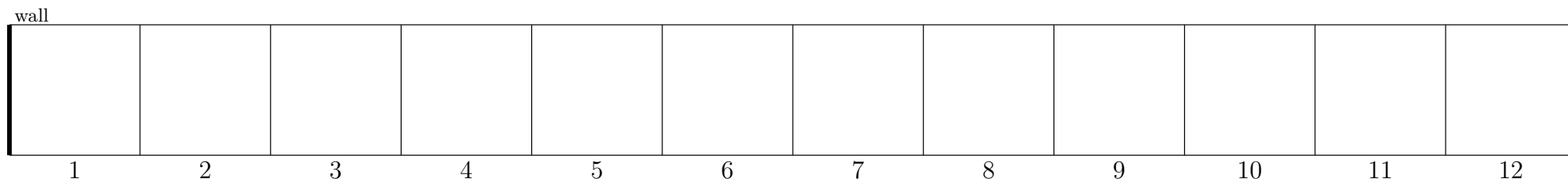
Winner:

Start 6



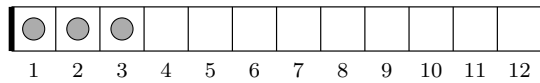
Winner:

Play here



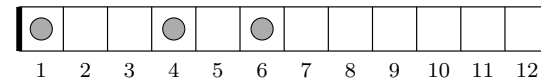
Problem 2 (continued): Play these three-coin starts. Find a rule for the starts where the second player can always win.

Start 7



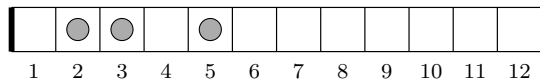
Winner: _____

Start 8



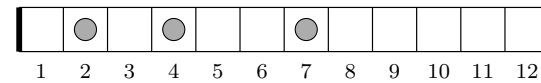
Winner: _____

Start 9



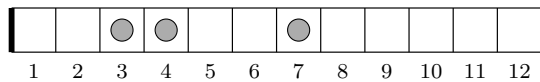
Winner: _____

Start 10



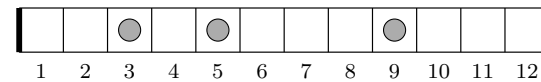
Winner: _____

Start 11



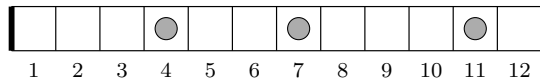
Winner: _____

Start 12



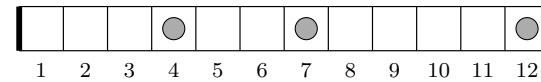
Winner: _____

Start 13



Winner: _____

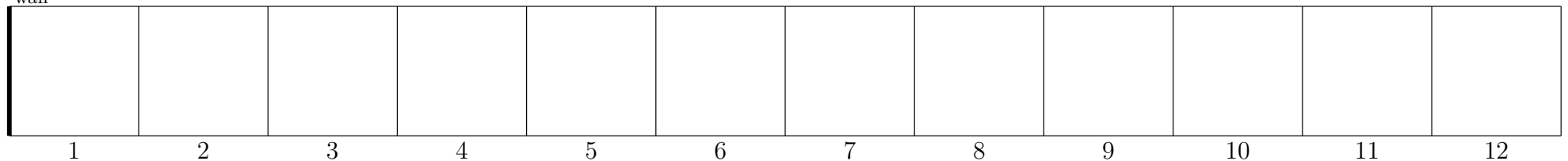
Start 14



Winner: _____

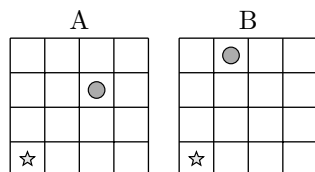
Rule: _____

Play here
wall

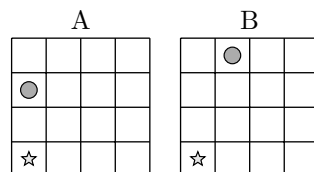


On your turn, choose one board. Slide its token one or more squares left or down. The player who leaves both tokens on their stars wins.

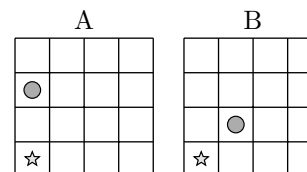
Before



A moves left 2 squares

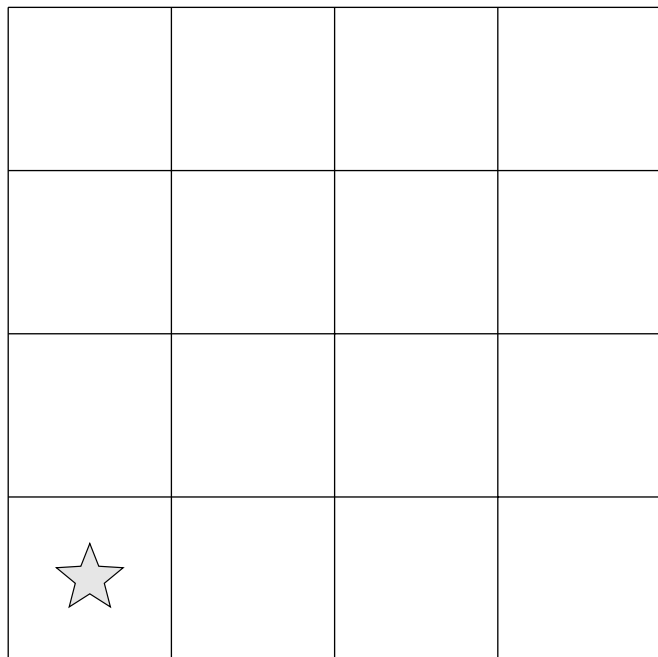


B moves down 2 squares

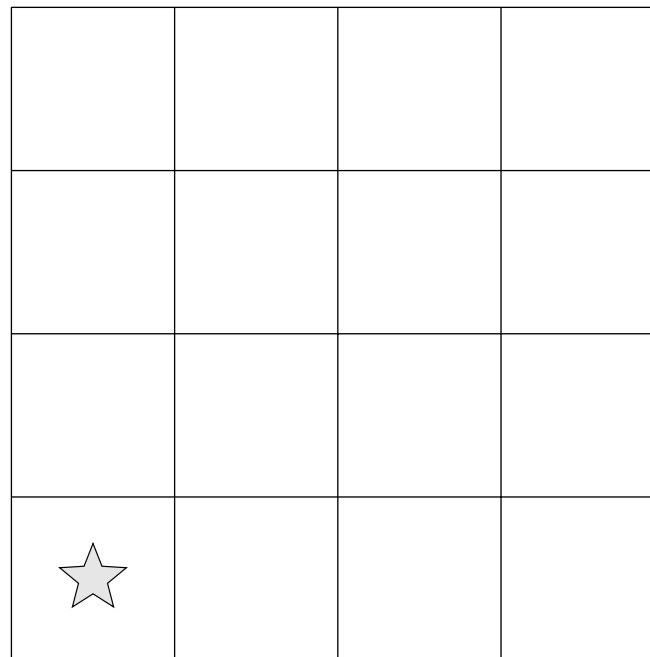


Problem 3: Play the pairs of starts on the next page. Which pairs let the first player always win? Invent another pair of different starts that lets the second player always win, even though each board alone lets the first player always win.

A: play here

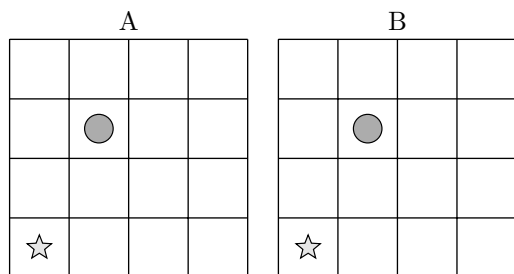


B: play here



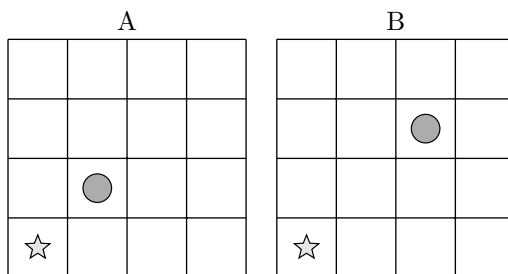
Problem 3 (continued): Which player can always win from each pair? Use the large boards to play.

Start 1



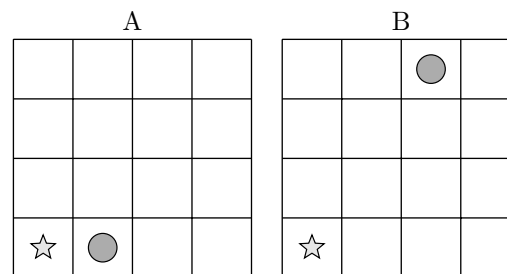
Winner: _____

Start 2



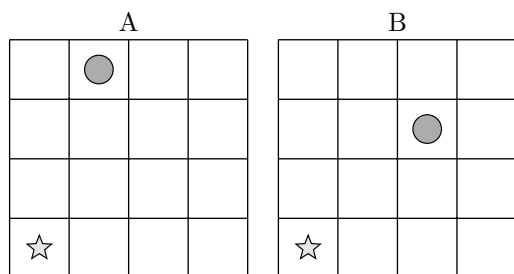
Winner: _____

Start 3



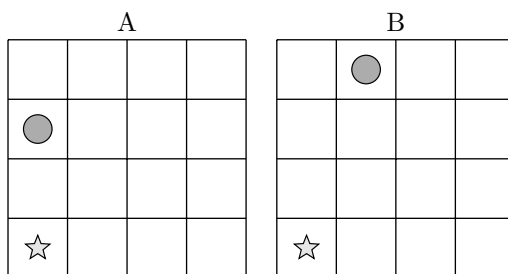
Winner: _____

Start 4



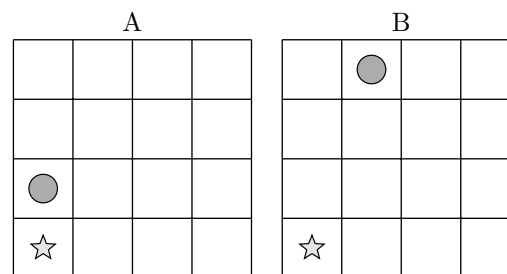
Winner: _____

Start 5



Winner: _____

Start 6



Winner: _____

Your pair:

